

Ing. Rick van Puijenbroek

Experienced analog/power electronics engineer.
I strive to enjoy life to the fullest. For me that not only means enjoying my private life, but also seeking challenges in a collaborative work environment and utilizing my diplomatic, mediator skills.

For design examples, refer to my portfolio at the end.

CONTACT

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6 May 1978

Married

Driver's license

COMPETENCES

Structured
Determined
Diplomatic
Result driven
Inquisitive
Communicative

SKILLS

Analog electronics
Power electronics
Low noise
High vacuum design
Architecture
Particle detectors
(x-ray, electrons)

SOFTWARE

LtSpice, TINA
Schematics & PCB CAD
Matlab
C++
MS Project

EDUCATION

Avans University of Applied Sciences – 's Hertogenbosch (Netherlands)
1995 – 1999 Bachelor of Science – electronics (telecommunication)

St. Odulphuslyceum Tilburg (Netherlands)
1991 – 1995 HAVO

PATENTS

June 14, 2013 Led tube for retrofitting in a fluorescent tube lighting fixture – EU pct/ib2013x/061444

EXPERIENCE

May 2021 – Present **Senior R&D electronic hardware engineer**
ThermoFisher Scientific – Image Acquisition & Detection group – Eindhoven (Netherlands)

- Designed proof-of-concept (POC) electronics for particle detector modules for high-end electron microscopy, based upon ASIC image sensors and silicon drift detectors. 20ppm accuracy, ≈100dB SNR
- Power electronics design, e.g. 200V low noise fast transient response and full bridge GaN stage for thermoelectric cooler
- System level design, specifications & interface requirements
- Alignment with other disciplines (firmware, mechanics, etc.)
- Lead co-development with outsourcing partners. ASIC image sensors, electronics (HDI PCB flex-rigid), firmware, mechatronics, EMC.
- Lead 8D investigations on 3rd line customer support issues
- Technical lead. Guide the outsourcing from contract (planning/cost), through requirements, design, integration testing up to release for manufacturing.
- Consulting and guiding conceptual design process of new detector modules
- Coaching junior colleagues

Keywords:
Architecture, system engineering, requirements management, technical lead, coaching, low noise, power/analog/digital electronics, SMPS, low noise, high voltage, PCB design, ASIC, silicon drift detectors, CMOS image sensors, design, simulation, qualification, integration, thermal, reliability, ultra-high vacuum, EMC, mechatronics, CE/UL

June 2016 – April 2021 **Senior R&D electronic hardware engineer**
Heidenhain motion control – R&D department – Eindhoven (Netherlands)

- Developed proof-of-concept electronics for future MagLev stage
- Designed low noise class-H amplifier (106dB SNR, 4,5kW$_{PK}$) using GaN MOSFETs
- Improved existing designs of low noise sensors and motor amplifiers.
- Alignment with other disciplines (firmware, mechanics, etc.)
- Introduced strategy for EMC compliance of modules and sub-assemblies.
- Involved in system architecture design of linear motors, motion systems, drive train electronics, metrology, connectivity.

HOBBIES

Hiking
Traveling
Good food
Culture

LANGUAGES

Dutch ★★★★★
English ★★★★★
German ★★★
French ★
Spanish ★

EXPERIENCE (CONTINUED)

Keywords:
Architecture, system engineering, power/analog/digital electronics, high voltage, SMPS, GaN, low noise, PCB design, sensors, actuators, mechatronics, design, simulation, qualification, integration, design for production, thermal, reliability, safety, EMC, CE/UL

May 2012 – May 2016 **Senior engineer (assigned via Yacht)**
ASML – EDEV, Image sensors & Measurements group – Eindhoven (Netherlands)
(Wafer alignment/levelling, Image alignment, Image quality)

- Designed extremely compact, low power, high dynamic range image sensor for metrology system to detect lens aberrations. Flex-rigid PCB located in wafer stage with custom made CMOS image sensor, low power FPGA, NTC, ADC's, SMPS's.
- Alignment with other disciplines (firmware, mechanics, etc.)
- Co-development with outsourcing partners: Caeleste, Jenoptik, VHDL
- Conceptual design, electronics & PCB design, qualification, integration
- Special challenge was limited 3D volume and high magnetic field from MagLev wafer stage.
- Assist project leader, guide the outsourcing. Cost planning estimate, requirements, design, integration testing, release for manufacturing.

Keywords:
Architecture, system engineering, requirements management, low noise, CMOS image sensors, low noise, low power, power/analog/digital electronics, sensors, design, qualification, integration, design for production, PCB, thermal, reliability, safety, EMC, CE/UL, planning

Sept 2014 – Mar 2016 **Mentor for ASML Scholarship Program**
ASML (part-time function next to engineering assignment) – Eindhoven (Netherlands)

- Act as ambassador to ASML and buddy to high potential students in the ASML Scholarship Program. www.asml.com/scholarship

Aug 2011 – Apr 2012 **Research scientist (assigned via Yacht)**
Philips Research – Solid State Lighting

- Research assignment into direct TL-tube replacement by LED-tube without needing to change the electronics in the fixture.
- Developed unique patented method to prevent electrical shock when replacing old style TL tubes by LED tubes.
- Designed proof-of-concept in limited 3D space. Qualified on TL drivers. Co-written worldwide patent application.
- Lobbied with IEEE board member on exact interpretation of safety standards.

Keywords:
Power/analog electronics, high voltage, design, simulation, qualification, reliability, safety, UL/CE, EMC, magnetics, thermal

EXPERIENCE (CONTINUED)

Dec 2010 – July 2011 **Power electronics designer (assigned via Yacht)**
Philips Lighting – eHID – Eindhoven (Netherlands)

- Redesigned resonant LLC HID driver with digital DALI interface (315W)
- Determine requirements, research into HID Philips proprietary knowledge, redesign existing driver including mechanical integration. Project cancelled during design phase.

Keywords:
Power/analog/digital electronics, PCB design, high voltage, mechanics, design, qualification, integration, design for production. Medical and commercial customers. CE/UL approbation, EMC/thermal design

2007 – 2010 **Analog designer / Project coordinator**
Wing B.V. – Eindhoven (Netherlands)

- Designed mixed signal electronics for various mechatronic applications
- Requirements & interface definition, planning, cost estimate, write statement of work
- (Conceptual) design, electronics & PCB design, qualification, integration
- Software designed by colleague in close collaboration
- Designed to various CE standards. E.g. intra-tank military handheld communication device. Electronics for medical blood analysis device with X-Y-Z motion stage, temperature-controlled chambers and CE compliance. Design involved FPGA, SMPS, motor stages, TEC drivers and assist in integration testing at customer.
- Other examples are wireless, low power, handheld (Li-Ion) sensors for temperature, light or strain gauges for automotive and military applications. Often complex flex-rigid PCBs where collaboration is required with mechanical designer of enclosures. Qualification involves pre-compliance CE testing and full compliance testing at notified body.
- Coaching junior colleagues

Keywords:
Power/analog/digital electronics, PCB design, planning, requirements management, mechatronics, qualification, integration, design for production, low power battery applications, wireless communication, sensors. Medical, military and commercial customers. CE approbation, EMC/thermal design.

2004 – 2007 **Power electronics designer**
Emotron A.G. (Sweden, Dutch R&D department) – Eindhoven (Netherlands)

- Designing, testing high power IGBT stages in new family AC drives (690V / 18.000kW).
- Designing, testing and integration of 400A choke and capacitor bank, in conjunction with mechanical design, sub-manufacturers, in-house production facilities and R&D site in Sweden.
- Reliability design & testing
- CE safety and EMC pre-compliance testing
- Challenges are cost, 3D space, CE/UL standards, thermal performance and lifetime expectancy.
- Responsible for quality improvement of design processes and documentation of Dutch R&D site, in collaboration with counterpart at site in Sweden.

Keywords:
Planning, power/analog/digital electronics, high voltage, mechanics, simulation, qualification, integration, design for production, voltage/current/temperature sensors. CE/UL approbation, EMC/thermal design.
Also active as assistant technical project leader for project leader based in Sweden.

EXPERIENCE (CONTINUED)

1999 – 2004 **Analog designer / Power electronics designer**
Prodrive B.V. – Eindhoven (Netherlands)

- Designed mixed signal electronics for various mechatronic applications
- Requirements & interface definition, planning, cost estimate
- Designed, tested, integrated at ASML high power, low noise AC/DC converters (48V / 1,2kW)
- High performance mixed signal designs, E.g. video digitization & processing for medical X-ray scanner, up to CE/UL approbation. Including FPGA, ADCs, SMPS, PCB design, design for medium volume production in-house and collaboration with mechanical designer of enclosures.
- Assisted in design of 10kW linear magnetic motor driver.

Keywords:
Power/analog/digital electronics, PCB design, high voltage, simulation, qualification, integration, high reliability, UL, CE, EMC, magnetics, thermal, lifetime

COURSES

2024 Introduction to Charged Particle Optics – Delft University of Technology
2022 CMOS imagers for electron microscopy workshop by Albert Theuwissen
2022 Basics & design principles for ultra-clean vacuum – High Tech Institute Eindhoven
2022 Systems Engineering – PPI
2019 Mechatronics system design – part 2 – High Tech Institute Eindhoven
2019 Consultancy skills – AOO
2018 Electromagnetic Compatibility EMC – PAOTM – Eindhoven University of Technology
2017 Mechatronics system design – part 1 – High Tech Institute Eindhoven
2016 Requirements writing workshop – ASML internal
2016 TI Precision labs – OpAmps
2014 Introduction to CMOS image sensors by Albert Theuwissen
2014 PCB technologies & design rules – ASML internal
2013 Masterclass part I - ILIAS and pupil scans – ASML internal
2012 Matlab fundamentals - MathWorks
2011 SIP training: coaching skills – Yacht internal
2011 Training projectmatig werken – Yacht internal
2011 SIP training: commercial skills – Yacht internal
2009 Knowledge Exchange Program - CE legislation and EMC-aware designing – Bicon EMC laboratories
2006 IR thermograph – High Tech Institute Eindhoven
2006 VCA basisveiligheid – Emotron internal
2006 Structured project based design approach – High Tech Institute Eindhoven
- Multiple Power electronic seminars
- Multiple CE/EMC/UL seminars

AREAS OF EXPERTISE

- Analog electronics (low noise/power)
- Particle detectors (silicon drift detectors, electron CMOS image ASICs)
- Power electronics (SMPS mW's to MW's, motor drives, lighting)
- Digital electronics (hardware implementation, no programming)
- High vacuum compatibility
- Technical project management (planning, V-model, structured design)
- Medical (x-ray scanner, brachy therapy)
- Military
- CE/UL/EMC approbation
- Wireless (ZigBee, NFC)

SKILLS

Analog electronics design

- Low noise/power design (analog-to-digital conversion in vacuum)
- ADC, DAC, JFET/OpAmp amplifiers
- Video, PLL, CMOS imaging, silicon drift detectors
- Sensors: strain gauges, thermocouple, piezo/ultrasonic transducers, etc.
- Battery chargers (NiMH, LiPo, LiIon, charging by solar cell or inductive (wireless energy))
- GPS, ZigBee and NFC transceivers (based upon reference design)

Power electronics design (SMPS)

- Lower power, high efficiency step-up/down (soft switching, 10mW to 2kW)
- High power variable frequency inverters (up to 1500kW)
- Lighting drivers (LED, TL, eHID)
- Motor drivers (class-H)

Digital electronic design

- SOC, FPGA design + peripherals (hardware part only)
- Interfaces (Ethernet, aSPI, RS-485, DALI, etc.)

PCB design

- Multi-layer flex-rigid, HDI, blind/buried via's, wire bonding
- Design for high vacuum environment

Magnetics design

- SMPS transformers/inductors
- 300Hz chokes

Thermal design

- Calculation of thermal design/lifetime expectancy
- Measurements using NTC/thermocouples, IR camera
- Use of environmental test laboratories

CE / UL

- Interpretation and design according to standards (Medical, EMC, Safety, R&TTE)
- Testing to standards (pre-compliance and full compliance using test labs)
- Consultancy to customers and colleagues
- Compiling technical construction dossier

Structured design approach

- Requirements management
- Proof of concept, planning, specification, design, qualification, integration, production support
- Coach junior colleagues
- Clear communication and documentation

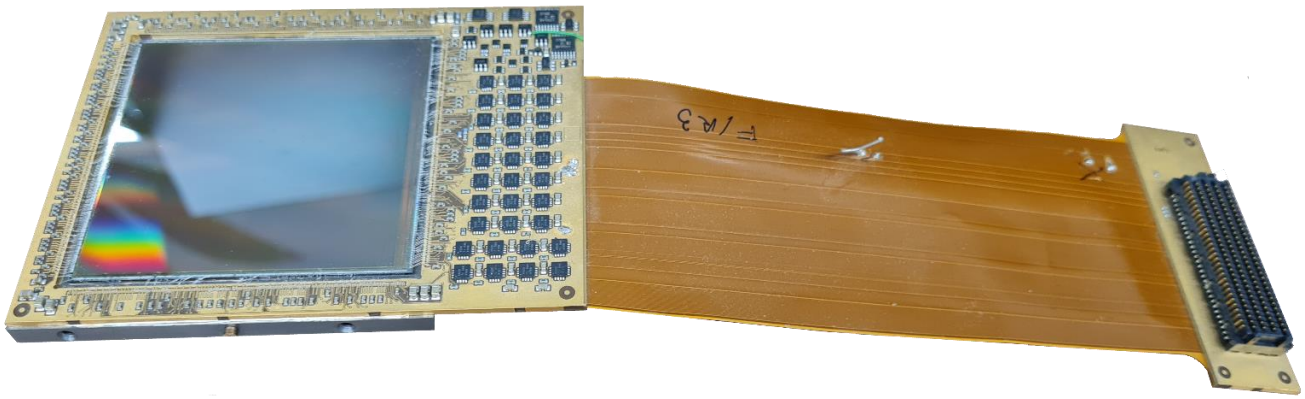
PORTFOLIO

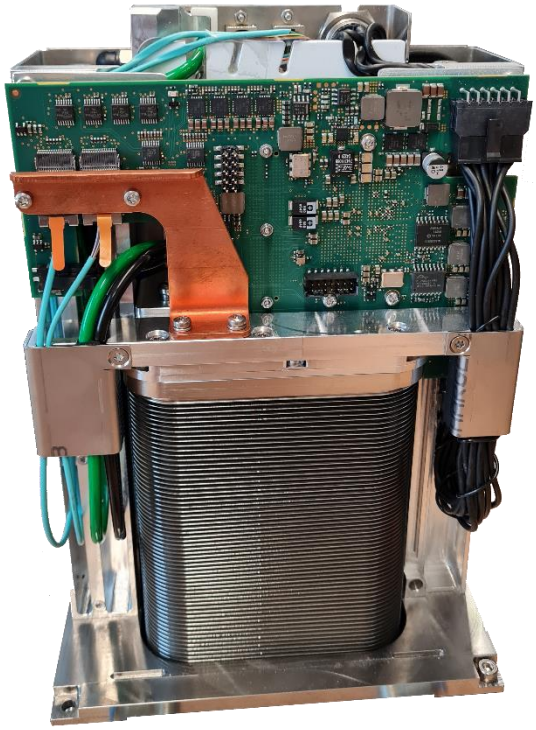
ThermoFisher Scientific

X-ray, direct & indirect electron detection cameras for Transmission Electron Microscopes.

My activities:

- Pre-study & conceptual system design with other disciplines (proof of concept)
- System & architecture design and alignment with machine interface & requirements
- Co-development with outsourcing partners
- Support integration activities (test plan & execution)
- Support project leader (planning, effort, cost, contract)
- Guide outsourcing





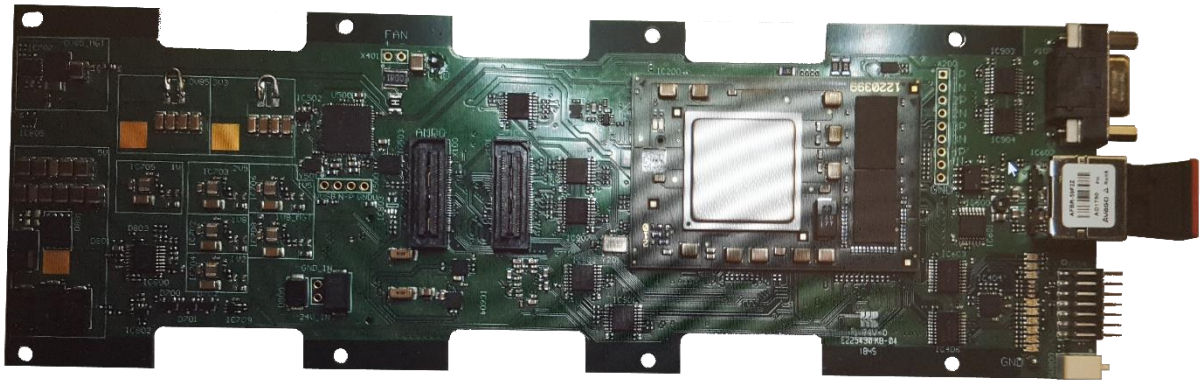


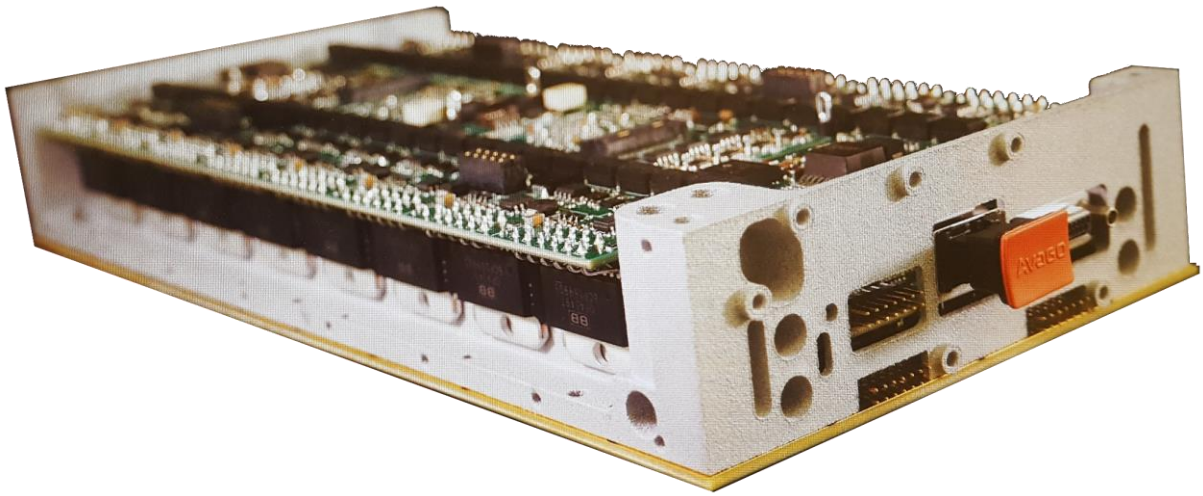
Heidenhain

My activities:

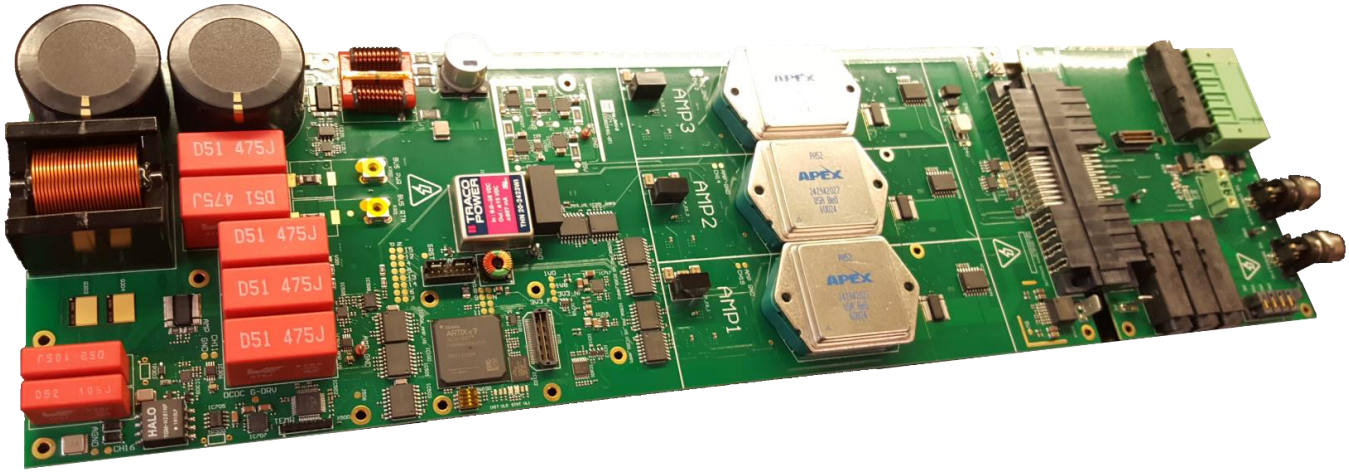
- Pre-study & conceptual system design with other disciplines (proof of concept)
- System & architecture design and alignment with machine interface & requirements
- Full electronics design from scratch, inc. PCB design, simulation (Spice/Matlab), control loop
- Close collaboration with digital designer
- Integration into machine (test plan & execution)

<u>16 channel motion platform, 50nm position accuracy, ZynQ SoC:</u>





<u>200V/20A/4kW class-H 3 channel motor drive (GaN) for maglev stage:</u>

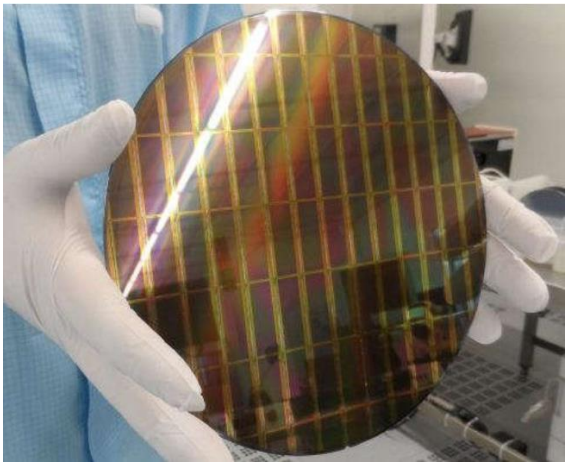


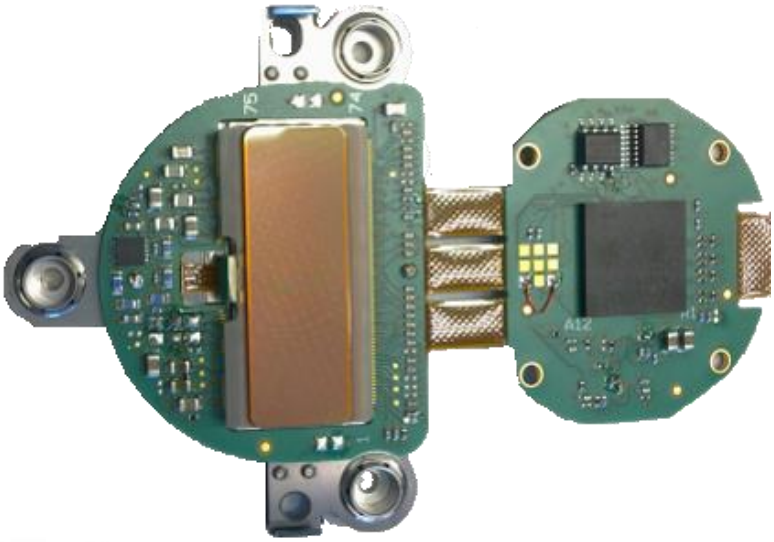
ASML

PARIS sensor in wafer stage, detects lens aberrations & reticle alignment:

My activities:

- Full electronics design, except firmware, with other disciplines
- Concept design and alignment with machine interface & requirements
- Co-development with outsourcing partners
- Integration & support manufacturing issues at outsourcing partners
- Support project leader. Guide the outsourcing







ASML

Emotron
530kW 690V_{AC} variable frequency drive:

My activities:

- Design & test IGBT power stages, DC choke, capacitor bank
- Reliability design & testing
- UL/CE safety and EMC pre-compliance testing
- Support technical lead in Sweden R&D site



